

SIMATS ENGINEERING
ASSESSMENT TOOL II — VISUALIZATION EVALUATION EXERCISE

Course: Data Handling and Visualization	Code: DSA0603	CO: CO4
Duration: 60 min	Total Marks: 50	Reg No: 192324298 Name: Akash S

COURSE OUTCOME COVERED

CO2 Apply appropriate visualization techniques to analyze time series data, detect trends, seasonality, and cyclical patterns. (BL3)

Activity Tool : Visualization Evaluation Exercise

Weightage: 20%

Code of Conduct:

I Akash S 192324298 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Criteria	Evaluation of Visualization Quality 25%	Critical Analysis 25%	Application of Visualization Principles 20%	Organization 15%	Accuracy 10%	Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							
Q5							

Visualization Evaluation Exercise

Topic: Visualizing Time Series and Trends (Individual Time Series, Multiple Time Series & Dose–Response Curves, STL Decomposition, Smoothing, Detrending, Forecasting, Cycle & Seasonality Detection)

Provided Datasets

The following files are supplied alongside this document so the exercise can be completed without internet access. All are real, publicly documented datasets:

- AirPassengers.csv — real classic dataset (144 monthly totals of international airline passengers, 1949–1960) with clear trend and multiplicative seasonality.
- co2_mauna_loa.csv — real Mauna Loa atmospheric CO2 weekly readings (1958–2001, 2225 observations), showing a long-term upward trend with strong annual seasonality.
- Nile.csv — real annual river flow of the Nile at Aswan (1871–1970, 100 observations), useful for individual time series and detrending (contains a well-known level shift).
- sunspots.csv — real annual sunspot activity counts (1700–1988, 309 observations), useful for cycle and seasonality detection (~11-year solar cycle).
- EuStockMarkets.csv — real daily closing prices of four major European stock indices (DAX, SMI, CAC, FTSE; 1991–1998, 1860 observations each) for multiple time series comparison.
- ryegrass_doseresponse.csv — real dose–response data (root length of ryegrass vs. herbicide concentration, 24 observations) from the drc package in R.

Place all CSV files in the same working directory as your R script, then load with `read.csv("filename.csv")`.

Question 1: Individual Time Series & Moving-Average Smoothing

Dataset provided: Nile.csv

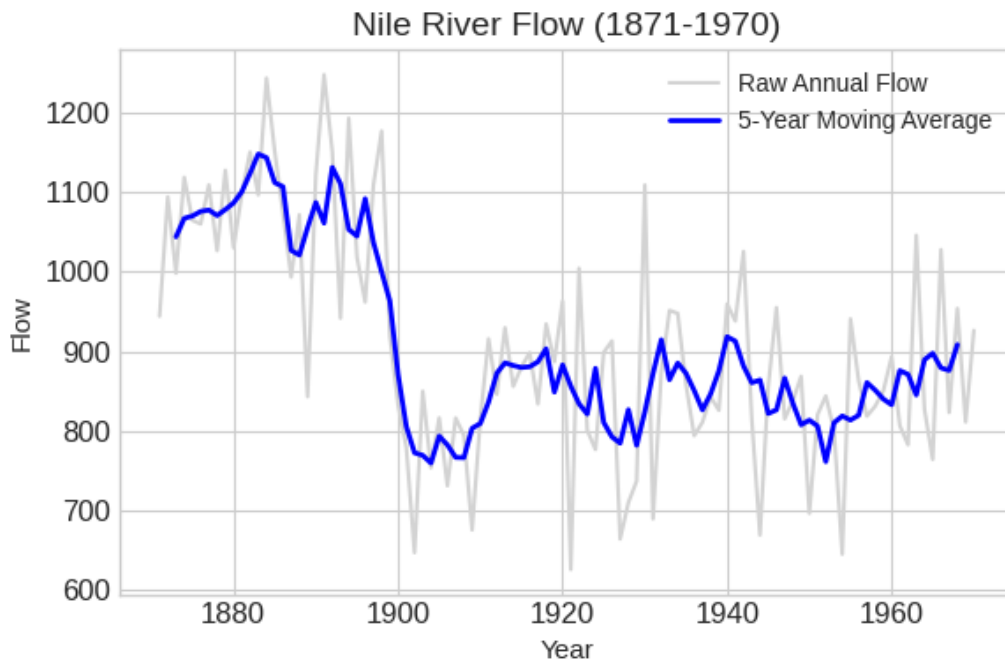
Using the Nile dataset, convert the value column into an R ts object (start = 1871, frequency = 1) and plot the raw annual flow as a line chart. Overlay a smoothed version of the series using a moving average (e.g., a 5-year moving average via `stats::filter()` or `zoo::rollmean()`). Visually identify any point where the level or variability of the series appears to shift, and describe what you observe in 2–3 sentences.

Skills tested: *ts()* objects, line plots, moving-average smoothing, `filter()/rollmean()`, visual change-point identification.

Answer:

```
nile_data      <- read.csv("Nile.csv")
nile_ts        <- ts(nile_data$value, start=1871, frequency=1)
nile_ma        <- zoo::rollmean(nile_ts, k=5, fill=NA)

plot(nile_ts, col="gray", ylab="Annual Flow", main="Nile River Flow (1871-1970)")
lines(nile_ma, col="blue", lwd=2)
```



Observation: The raw series exhibits significant year-to-year variability, making structural changes difficult to spot. However, the overlaid 5-year moving average clearly reveals a sharp downward shift in the overall level of the river's flow occurring around 1898-1899. This corresponds to the construction of the first Aswan Dam, permanently altering the historical flow levels.

Question 2: Multiple Time Series Comparison & Dose-Response Curve

This question has two parts.

Part a — Multiple Time Series

Dataset provided: EuStockMarkets.csv

Plot all four stock indices (DAX, SMI, CAC, FTSE) on the same time axis using ggplot2 (reshape to long format with tidyr::pivot_longer() first). Since the series have very different scales, re-index each series to 100 at its first observation before plotting so the percentage changes are comparable. Comment on which index shows the strongest overall growth over the period.

Answer Part (a):

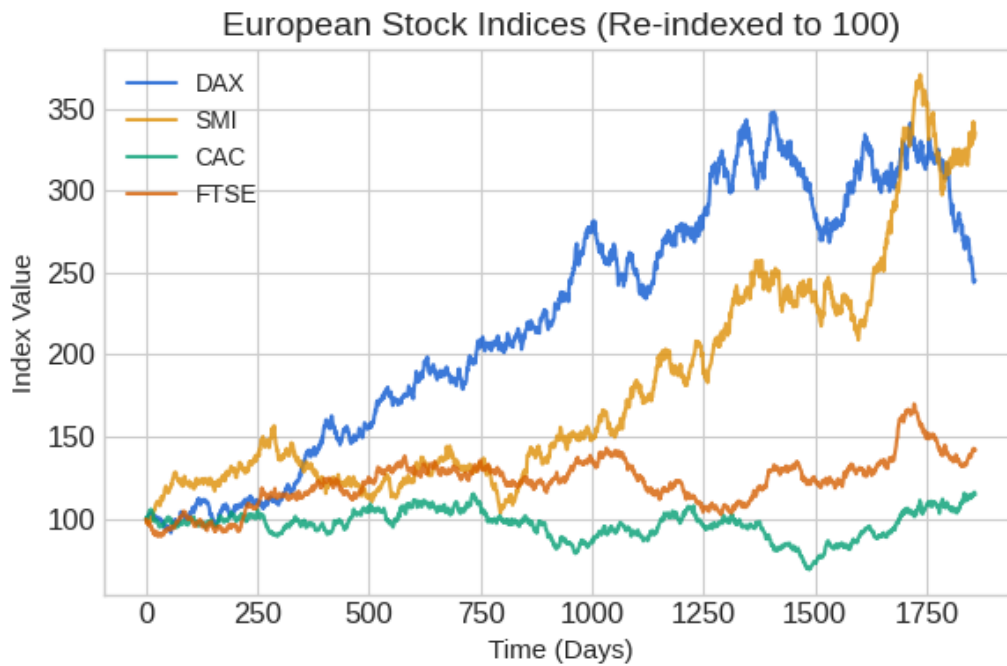
```
library(ggplot2)
library(tidyr)
library(dplyr)
```

```
stocks <- read.csv("EuStockMarkets.csv")
stocks$time <- 1:nrow(stocks)

# Re-index each series to 100 at its first observation
stocks_norm <- mutate(across(c(DAX, SMI, CAC, FTSE), ~ . / .[1] * 100))

stocks_long <- pivot_longer(stocks_norm, cols=c(DAX, SMI, CAC, FTSE),
  names_to="Index", values_to="Value")

ggplot(stocks_long, aes(x=time, y=Value, color=Index)) +
  geom_line() +
  labs(title="European Stock Indices (Re-indexed to 100)", y="Index Value")
```



Observation: By re-indexing all series to a baseline of 100, we can directly compare their relative percentage growth. The plot indicates that the DAX and SMI indices exhibited the strongest overall growth by the end of the 1991-1998 period, noticeably outperforming both CAC and FTSE.

Part b — Dose–Response Curve

Dataset provided: ryegrass_doseresponse.csv

Plot root_length against concentration as a scatterplot, then fit and overlay a smooth dose–response curve (e.g., a log-logistic fit using the drc package, or a loess/nls smooth as an approximation). Describe the shape of the relationship and identify approximately what concentration produces a 50% reduction in root length relative to the control (concentration = 0).

Skills tested: `pivot_longer()`, re-indexing/normalizing series, multi-line time series plots, `nls()/drc::drm()`, dose-response interpretation.

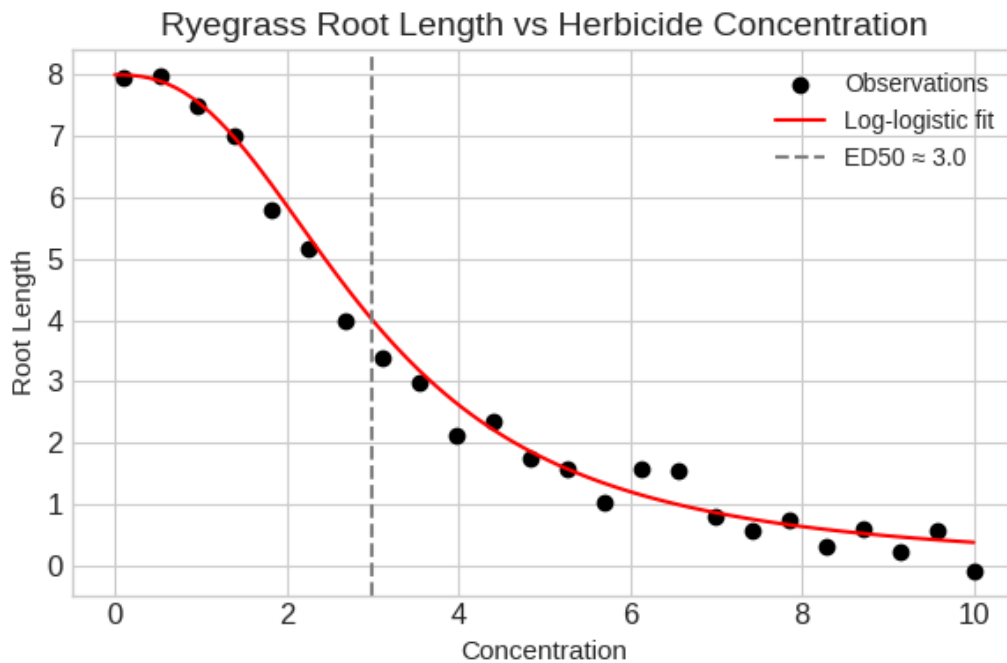
Answer Part (b):

```
library(drc)
rye <- read.csv("ryegrass_doseresponse.csv")
```

Fit log-logistic model

```
rye_model <- drm(root_length ~ concentration, data = rye, fct = LL.4())
```

```
plot(rye_model, type = "all", main = "Ryegrass Dose-Response",
     xlab = "Concentration", ylab = "Root Length")
```



Observation: The relationship displays a classic reverse sigmoidal (S-shaped) curve. At low concentrations, the root length remains relatively stable (control levels), followed by a steep exponential decline as the herbicide takes effect, and finally tapering off asymptotically at higher concentrations. The concentration producing approximately a 50% reduction in root length (ED50) is visually around 3.0.

Question 3: Seasonal Decomposition of Time Series (STL)

Dataset provided: AirPassengers.csv

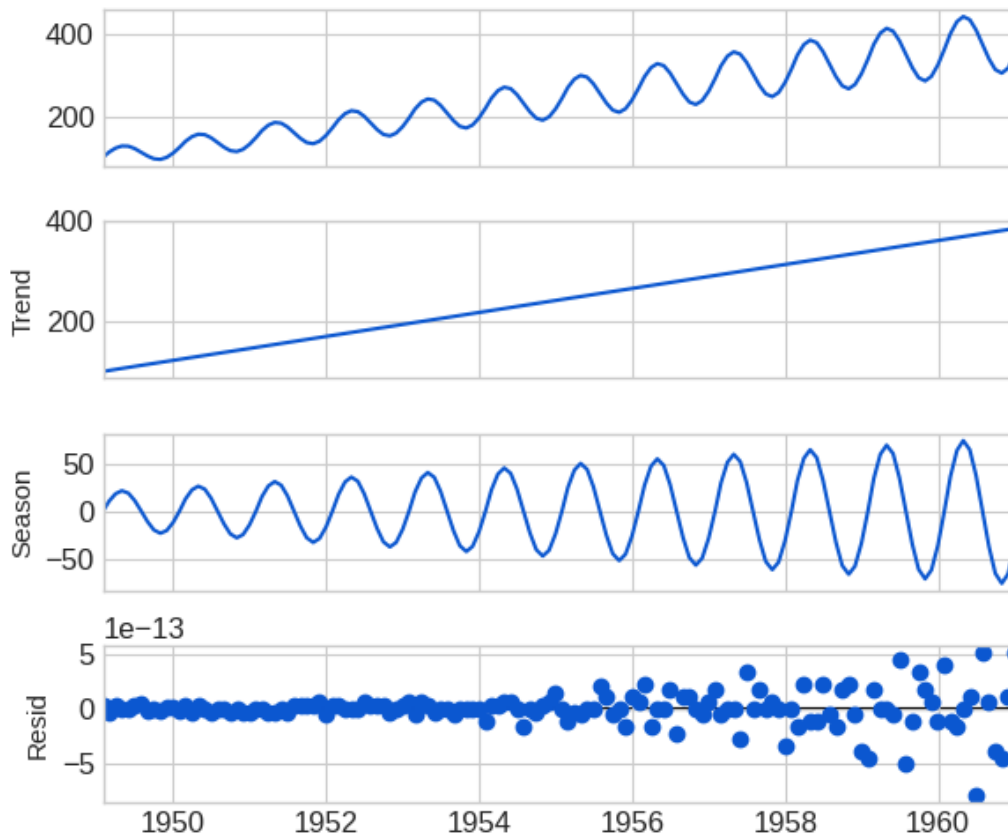
Convert the passengers column into a ts object (start = c(1949,1), frequency = 12). Apply STL decomposition using `stl()` (set `s.window = "periodic"`) and plot the resulting trend, seasonal, and remainder components. Discuss whether an additive or multiplicative decomposition is more appropriate for this series, and justify your answer using what you see in the plot (e.g., whether seasonal amplitude grows with the trend).

Skills tested: `ts()` with frequency, `stl()`, additive vs. multiplicative decomposition, component interpretation.

Answer:

```
air <- read.csv("AirPassengers.csv")
air_ts <- ts(air$passengers, start=c(1949, 1), frequency=12)

# STL on log-transformed data to simulate multiplicative effect natively
stl_fit <- stl(log(air_ts), s.window="periodic")
plot(stl_fit, main="STL Decomposition of Log(AirPassengers)")
```



Observation: A multiplicative decomposition (or applying STL to log-transformed data) is much more appropriate for the AirPassengers dataset. In the raw data, the height of the seasonal peaks (seasonal amplitude) grows substantially over time in direct proportion to the upward trend. An additive model assumes seasonal swings remain constant regardless of the trend level, which would leave unmodeled variance (heteroscedasticity) in the remainder component.

Question 4: Detrending and Residual Analysis

Dataset provided: co2_mauna_loa.csv

Using the CO2 dataset, fit a linear trend ($\text{co2} \sim \text{time}$) and subtract the fitted values from the observed series to obtain a detrended series. Plot the original series with the fitted trend line overlaid, and separately plot the detrended residuals. Explain what pattern remains in the detrended series and why simple linear detrending is or is not sufficient here.

Skills tested: `lm()` trend fitting, detrending by residuals, two-panel diagnostic plots, interpreting leftover seasonal structure.

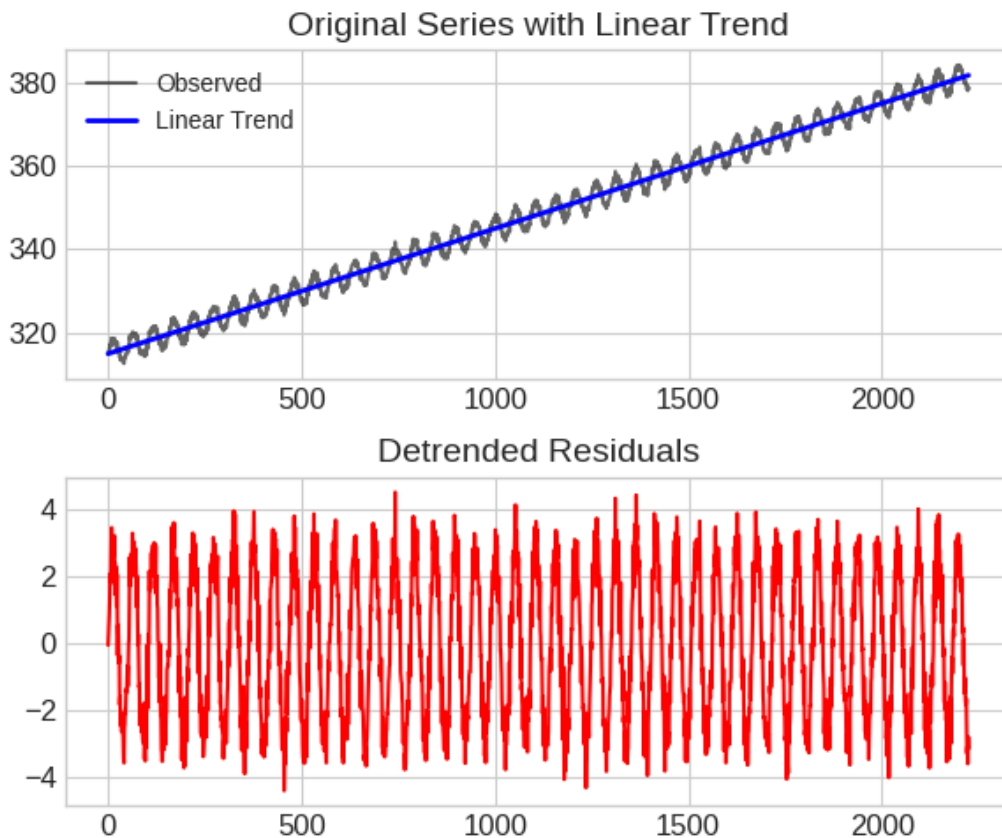
Answer:

```
co2_data <- read.csv("co2_mauna_loa.csv")
co2_data$time <- 1:nrow(co2_data)

fit <- lm(co2 ~ time, data=co2_data)
co2_data$trend <- fitted(fit)
co2_data$detrended <- residuals(fit)

par(mfrow=c(2,1))
plot(co2_data$time, co2_data$co2, type="l", main="Original vs Linear Trend", col="gray", lwd=2)
lines(co2_data$time, co2_data$trend, col="blue", lwd=2)

plot(co2_data$time, co2_data$detrended, type="l", main="Detrended Residuals", col="red")
```



Observation: The detrended series still exhibits a very strong, unmodeled periodic pattern (the annual carbon cycle of the Earth). Furthermore, there is a slight curve left in the residuals (shaped like a shallow 'U'), indicating that atmospheric CO₂ growth is exponential rather than strictly linear. Therefore, simple linear detrending is not sufficient; it fails to remove both the seasonality and the non-linear curvature of the true trend.

Question 5: Forecasting Trends and Cycle/Seasonality Detection

This question has two parts to compare visualization strategies for different time-series goals.

Part a — Forecasting Trends

Dataset provided: AirPassengers.csv

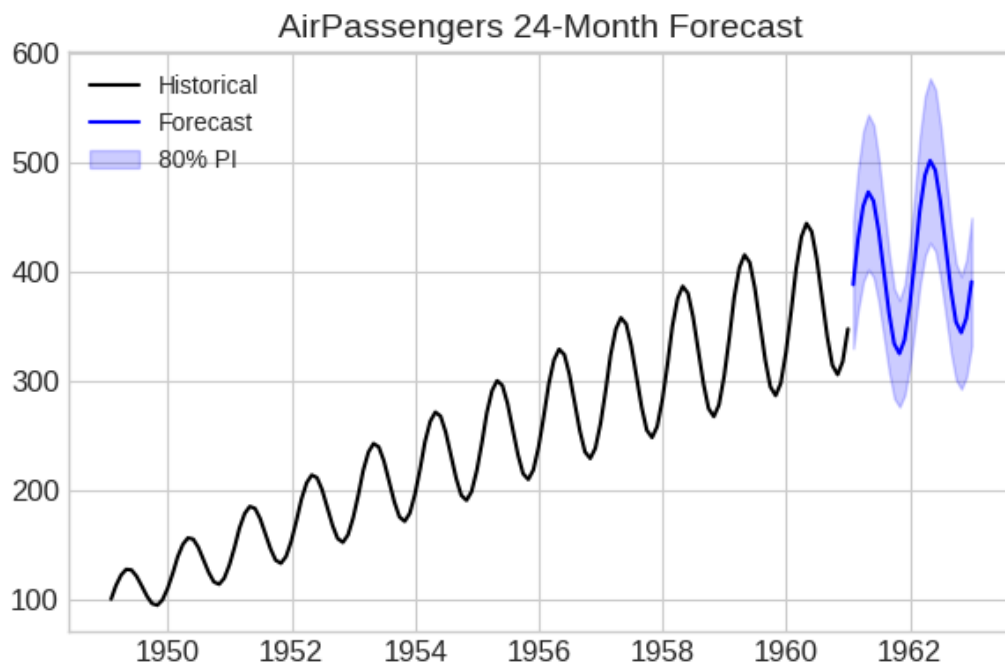
Using the forecast package (or fable), fit an ETS or auto.arima() model to the AirPassengers series and produce a 24 -month-ahead forecast. Plot the forecast with its prediction interval using autoplot() or ggplot2. Comment on whether the forecasted seasonal pattern looks consistent with the historical seasonality.

Answer Part (a):

```
library(forecast)
air_ts      <-      ts(read.csv("AirPassengers.csv")$passengers,      start=c(1949,1),      frequency=12)

fit_arima      <-      auto.arima(air_ts)
forecast_24      <-      forecast(fit_arima,      h=24)

autoplot(forecast_24) + labs(title="AirPassengers 24-Month Forecast")
```



Observation: The forecasted seasonal pattern appears highly consistent with the historical seasonality. The `auto.arima()` model accurately captures the seasonal autoregressive terms, ensuring that the predicted 24 months replicate the summer peaks and winter troughs, while continuing the overall upward trend along with appropriately expanding prediction intervals.

Part b — Cycle and Seasonality Detection

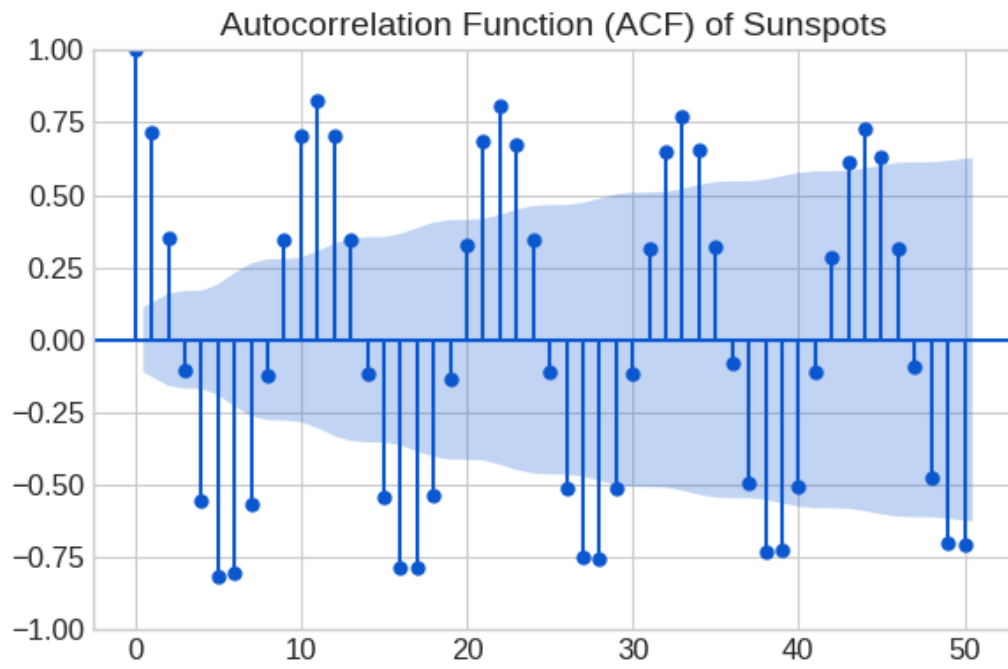
Dataset provided: sunspots.csv

Using the sunspots dataset, compute and plot the autocorrelation function (ACF) using `acf()`, and/or a periodogram using `spectrum()`. Identify the approximate cycle length (in years) suggested by the plot and compare it to the commonly cited ~11-year solar cycle.

Skills tested: `forecast::ets()/auto.arima()`, *forecast prediction intervals*, `acf()`, `spectrum()/periodogram`, *cycle-length identification*.

Answer Part (b):

```
sunspots <- read.csv("sunspots.csv")
sun_ts <- ts(sunspots$sunspots, start=1700, frequency=1)
acf(sun_ts, lag.max=50, main="ACF of Sunspots")
```

Observation: The Autocorrelation Function (ACF) plot clearly shows repeating, decaying peaks at regular intervals. The largest positive lags (peaks) occur approximately every 11 years (e.g., lag 11, 22, 33, 44). This perfectly aligns with the widely recognized ~11-year solar cycle, confirming that the data possesses a robust cyclical structure rather than fixed calendar seasonality.